# Debugging shapes and gradients

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

## One readable shape-and-gradient repair record.

Trace the first discrepancy,choose revealing inputs and verify the saved-state repair.

**Keep one habit:** W1–W3/W5 → selected new gradient → project repair ledger → changed-case return.

MAIN LESSON · SECTION 1

## Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from **Training** and validation curves. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from **Training** a small network. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- For this lesson, a dense layer stores one **output** unit per weight-matrix row. If an **input** has two features and the layer has three units, W has three rows and two columns, b has length 3, and one **input** x has length 2.

  Compute z for each unit, then apply ReLU to each component.

- Use W rows [1, 0], [0, 1], [1, -1] and b = [0, 0, 1]. For x = [1, 2], the scores are [1, 2, 0] and the activations are [1, 2, 0].

  A batch stores examples in rows: **input**s [[1, 2], [3, 1]] have **shape** 2 by 2, and **output**s have **shape** 2 by 3.

- Other software may use a transposed weight convention. The key is to record and follow one **contract**. An array with the right number of values can still be wrong if features or axes are swapped.

### Initialisation and reproducibility

**My remembered idea, example and correction**

### Training and validation curves

**My remembered idea, example and correction**

### Training a small network

**My remembered idea, example and correction**

### Lesson reference for workbook practice

| Object | Declared shape | Meaning |
| --- | --- | --- |
| One input x | Length 2 | Two features |
| Weights W | 3 by 2 | Three units each reading two features |
| Bias b | Length 3 | One bias per unit |
| Batch output | 2 by 3 | Two examples each with three activations |

---

MAIN LESSON · SECTION 5

## W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

### Complete task and evidence

- Main Wrows[1, 0]/[0, 1]/[1, −1], b[0, 0, 1], inputlength 2; batch 2 rows→outputs 2×3. x[1, 2]scores[1, 2, 0], zeroedb 3 scores[1, 2, −1], sameReLU; [0, 2]scores[0, 2, −1]versus[0, 2, −2], sameReLU; [3, 1]scores/activations[3, 1, 3]versus[3, 1, 2], exposeserror. Omitted term means replaceb 3 with0, notremoveitsslot.

  Separate picturey 2a+3c, target 4, old[1, 0], gradient[a, c]=[−8, −12]; rate.01 new[1.08, .12], y2.52, loss 2.1904. Diagnostic y2w+b, t3, old[1, 0], gradient[−4, −2], rate.05 subtract→loss.25, add→2.25. Finite values assumed; correct shapes and one loss decrease are incomplete checks.

- The layer expects W **shape** 3 by 2, b length 3 and x length 2. A submitted x has length 3. Should the last feature be silently ignored? State the failed **contract** and repair **process**.

> **My reasoning and evidence**

---

MAIN LESSON · SECTION 6

## W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Main Wrows[1, 0]/[0, 1]/[1, −1], b[0, 0, 1], inputlength 2; batch 2 rows→outputs 2×3. x[1, 2]scores[1, 2, 0], zeroedb 3 scores[1, 2, −1], sameReLU; [0, 2]scores[0, 2, −1]versus[0, 2, −2], sameReLU; [3, 1]scores/activations[3, 1, 3]versus[3, 1, 2], exposeserror. Omitted term means replaceb 3 with0, notremoveitsslot.

  Separate picturey 2a+3c, target 4, old[1, 0], gradient[a, c]=[−8, −12]; rate.01 new[1.08, .12], y2.52, loss 2.1904. Diagnostic y2w+b, t3, old[1, 0], gradient[−4, −2], rate.05 subtract→loss.25, add→2.25. Finite values assumed; correct shapes and one loss decrease are incomplete checks.

- Compute scores and activations for x = [3, 1] with the lesson layer, then omit the third bias. Identify the first discrepancy and the resulting activation error.

> **My reasoning and evidence**

---

MAIN LESSON · SECTION 7

## W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

### Complete task and evidence

- Main Wrows[1, 0]/[0, 1]/[1, −1], b[0, 0, 1], inputlength 2; batch 2 rows→outputs 2×3. x[1, 2]scores[1, 2, 0], zeroedb 3 scores[1, 2, −1], sameReLU; [0, 2]scores[0, 2, −1]versus[0, 2, −2], sameReLU; [3, 1]scores/activations[3, 1, 3]versus[3, 1, 2], exposeserror. Omitted term means replaceb 3 with0, notremoveitsslot.

  Separate picturey 2a+3c, target 4, old[1, 0], gradient[a, c]=[−8, −12]; rate.01 new[1.08, .12], y2.52, loss 2.1904. Diagnostic y2w+b, t3, old[1, 0], gradient[−4, −2], rate.05 subtract→loss.25, add→2.25. Finite values assumed; correct shapes and one loss decrease are incomplete checks.

- For the diagnostic model at [1, 0], gradient [-4, -2] and rate 0.05, calculate the correct new parameters and loss. Compare with adding the gradient step.

> **My reasoning and evidence**

---

MAIN LESSON · SECTION 9

## W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Main Wrows[1, 0]/[0, 1]/[1, –1], b[0, 0, 1], inputlength 2; batch 2 rows→outputs 2×3. x[1, 2]scores[1, 2, 0], zeroedb 3 scores[1, 2, –1], sameReLU; [0, 2]scores[0, 2, –1]versus[0, 2, –2], sameReLU; [3, 1]scores/activations[3, 1, 3]versus[3, 1, 2], exposeserror. Omitted term means replaceb 3 with0, notremoveitsslot.

  Separate picturey 2a+3c, target 4, old[1, 0], gradient[a, c]=[–8, –12]; rate.01 new[1.08, .12], y2.52, loss 2.1904. Diagnostic y2w+b, t3, old[1, 0], gradient[–4, –2], rate.05 subtract→loss.25, add→2.25. Finite values assumed; correct shapes and one loss decrease are incomplete checks.

- A program passes only the x = [1, 2] final-activation test. Propose one intermediate-value check, one malformed-**input** check and one additional **input** that exposes the bias omission.

> **My reasoning and evidence**

---

MAIN LESSON · SECTION 4

## Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Use y = 3a + c, target 2, loss (y - 2) squared, at [a, c] = [0, 1]. A program returns gradient [-2, -6] in order [a, c].

  Use symmetric step 0.1 to check each coordinate, then update both with the corrected gradient and rate 0.05. Report new parameters and loss.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Independent case: Debug a new gradient independently

> **My independent explanation, trace and limits**

---

MAIN LESSON · SECTION 10

## Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Add a labelled debugging **shape**s and gradients **evidence** sheet to Image-pattern Network Investigation. Show how you can check dimension **contract**s before computing a layer.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

# Feedback · Repair and return

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

## Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Use W2: Expose a hidden bias error for Debugging **shape**s and gradients. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

# Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- Use the worked case for Debugging **shape**s and gradients. Proposed explanation: "Matching **output shape** guarantees the computation". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

# Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- Use y = 3a + c, target 2, loss (y - 2) squared, at [a, c] = [0, 1]. A program returns gradient [-2, -6] in order [a, c].

  Use symmetric step 0.1 to check each coordinate, then update both with the corrected gradient and rate 0.05. Report new parameters and loss.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

# Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- Catch a gradient -order bug before the optimiser changes parameters.

- Toy y=2a+3c, target 4, start [a, c]=[1, 0]. Loss $(y-4)^2$. Program claims gradient [-12,-8] in order [a, c].

- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- What should a debugging record pair with every gradient value? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/18

# Debugging shapes and gradients

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.

### 1. Compute correct gradient

y = 2, residual -2: gradient [-8,-12], not [-12,-8].

### 2. Check coordinates

Step 0.1: a losses 3.24, 4.84 give -8; c losses 2.89, 5.29 give -12.

### 3. Apply correct update

At rate 0.01: new [a, c]=[1.08, 0.12]; y = 2.52, loss 2.1904.

**CHANGE ONE THING**

The program changes its declared order to [c, a] and returns [-12,-8].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

**My picture-based explanation and changed-case prediction**

▼ Reference · Project milestone

REFERENCE · SECTION 13

# Reference · Project milestone

> Keep this task’s inputs,rule and response together.

## Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

| Evidence to keep | My record |
| --- | --- |
| This week’s input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

# Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

## First supported question and optional hint

- Catch a gradient -order bug before the optimiser changes parameters.
- Which labelled step matches this detail? At rate 0.01: new [a, c]=[1.08, 0.12]; y = 2.52, loss 2.1904.
- 1. Compute correct gradient
- 2. Check coordinates
- 3. Apply correct update
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Apply correct update At rate 0.01: new [a, c]=[1.08, 0.12]; y = 2.52, loss 2.1904. Array **shape** and coordinate meaning both matter.

A vector with two entries can have the right length but assign each derivative to the wrong parameter. Label the entries a and c, then perturb one coordinate at a time to check their slopes.

Repairing the coordinate mapping is necessary before using the gradient to update the network.

## Changed-case question and optional hint

- Changed-case practice: The program changes its declared order to [c, a] and returns [-12,-8]. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. That vector now matches the declared coordinate order.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: That vector now matches the declared coordinate order. Gradient values are meaningful only when their parameter names and array positions agree.

## Repair and retain the record

Review Initialisation and reproducibility (ai-5/17). Goal: Trace how starting values affect a fixed **training** budget. Short practice: Does the matching final loss mean the parameter values are equal?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Training** and validation curves (ai-5/15). Goal: Compute comparable **training** and validation losses at saved checkpoints. Short practice: Can the original epoch 3 **training** loss 1 justify claiming best generalisation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Training** a small network (ai-5/12). Goal: Trace repeated forward loss gradient and update steps. Short practice: Which parameter changed, and which part was not trained?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Predict, run and debug**

**Keep your completed work**

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

## Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

## 01. Debugging shapes and gradients: the case

AI ACADEMY / AI-5/18 · 01 · LOOK AT THE FACTS

### Debugging shapes and gradients: the case

**The situation**

Catch a gradient -order bug before the optimiser changes parameters.

**Supplied evidence**

Toy y=2a+3c, target 4, start [a, c]=[1, 0]. Loss (y-4)². Program claims gradient [-12,-8] in order [a, c].
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

**By the end, I can explain debugging shapes and gradients using a traced example and a stated limitation.**

Follow the labelled picture with the matching explanation. Open the vector diagram

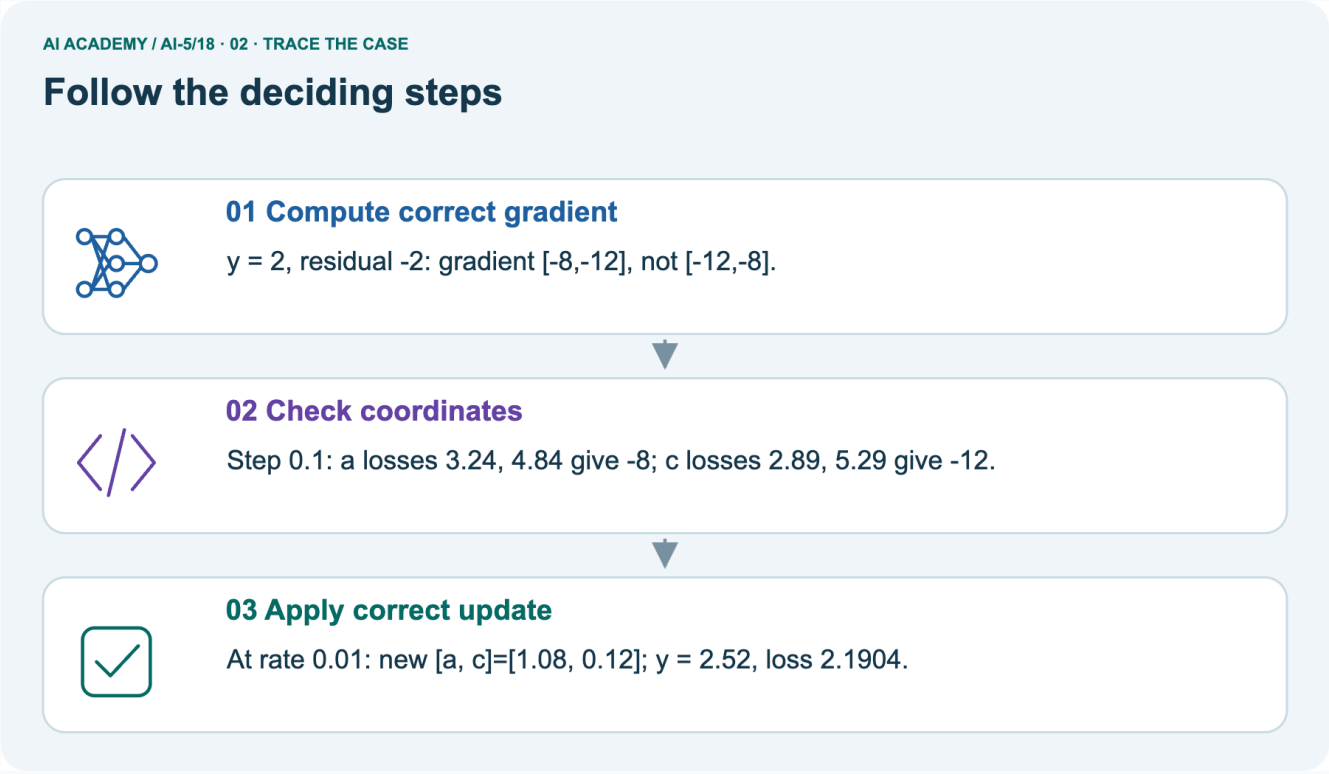
Catch a gradient -order bug before the optimiser changes parameters. Supplied facts: Toy y=2a+3c, target 4, start [a, c]=[1, 0]. Loss (y-4)². Program claims gradient [-12,-8] in order [a, c].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

**My picture explanation, deciding evidence and one limit**

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

Compute correct gradient: y = 2, residual -2: gradient [-8,-12], not [-12,-8]. → Check coordinates: Step 0.1: a losses 3.24, 4.84 give -8; c losses 2.89, 5.29 give -12. → Apply correct update: At rate 0.01: new [a, c]=[1.08, 0.12]; y = 2.52, loss 2.1904.

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

## 03. Repair from the first disagreement

AI ACADEMY / AI-5/18 · 03 · INSPECT THE LESSON MODEL

### Repair from the first disagreement

| Trace stage for x [1 2] | Expected result | Submitted result |
| --- | --- | --- |
| Input and dimensions | x length 2 W 3 by 2 b length 3 | Dimensions match |
| Scores before ReLU | [1, 2, 0] | [1, 2, -1] |
| Hidden activations | [1, 2, 0] | [1, 2, 0] |

The submitted third score omits bias 1. ReLU hides the error for this input, so matching final activations are not enough. At x = [3, 1], the same omission changes the third activation from 3 to 2.

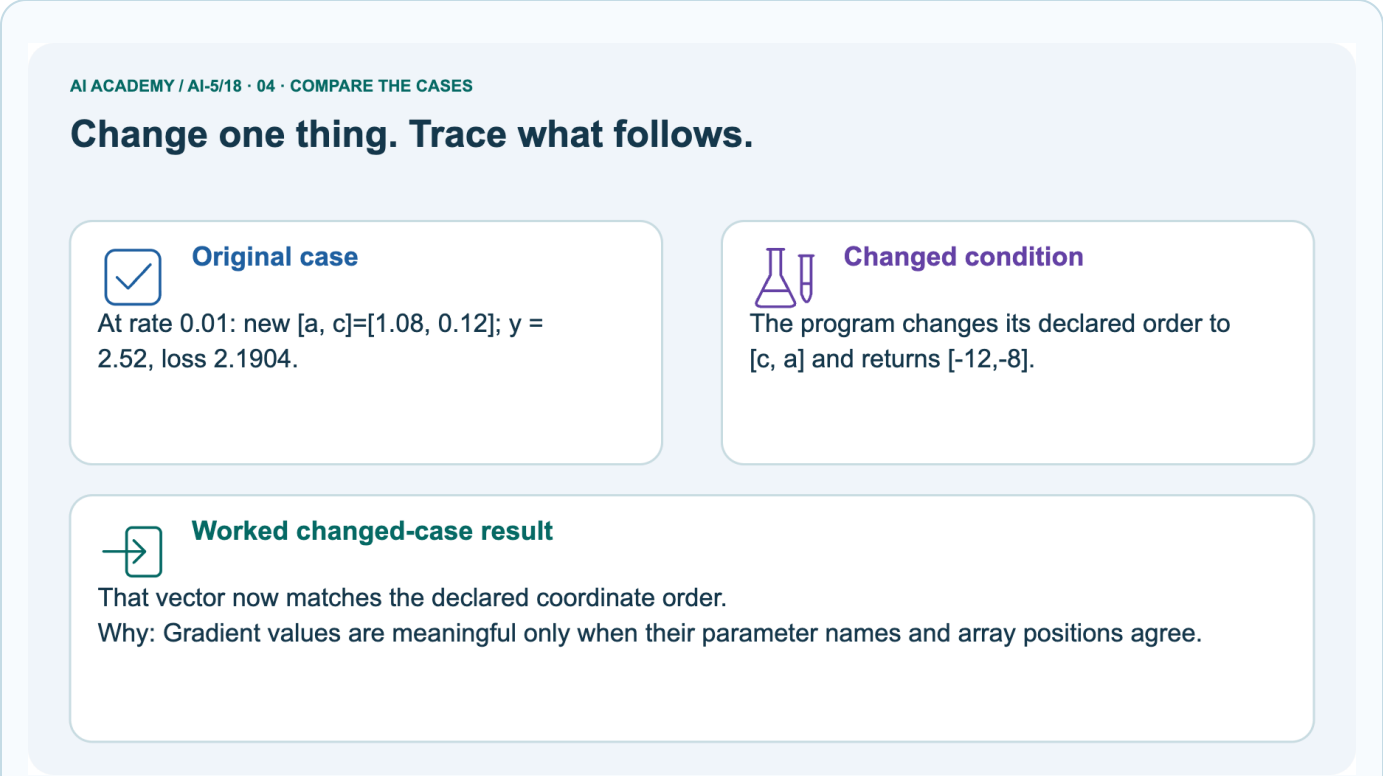
Follow the labelled picture with the matching explanation. Open the vector diagram

Repair from the first disagreement. **Input** and dimensions | x length 2 W 3 by 2 b length 3 | Dimensions match. Scores before ReLU | [1, 2, 0] | [1, 2, -1]. Hidden activations | [1, 2, 0] | [1, 2, 0]

Read the labels and values. Explain the relationship the model shows.

### My picture explanation, deciding evidence and one limit

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

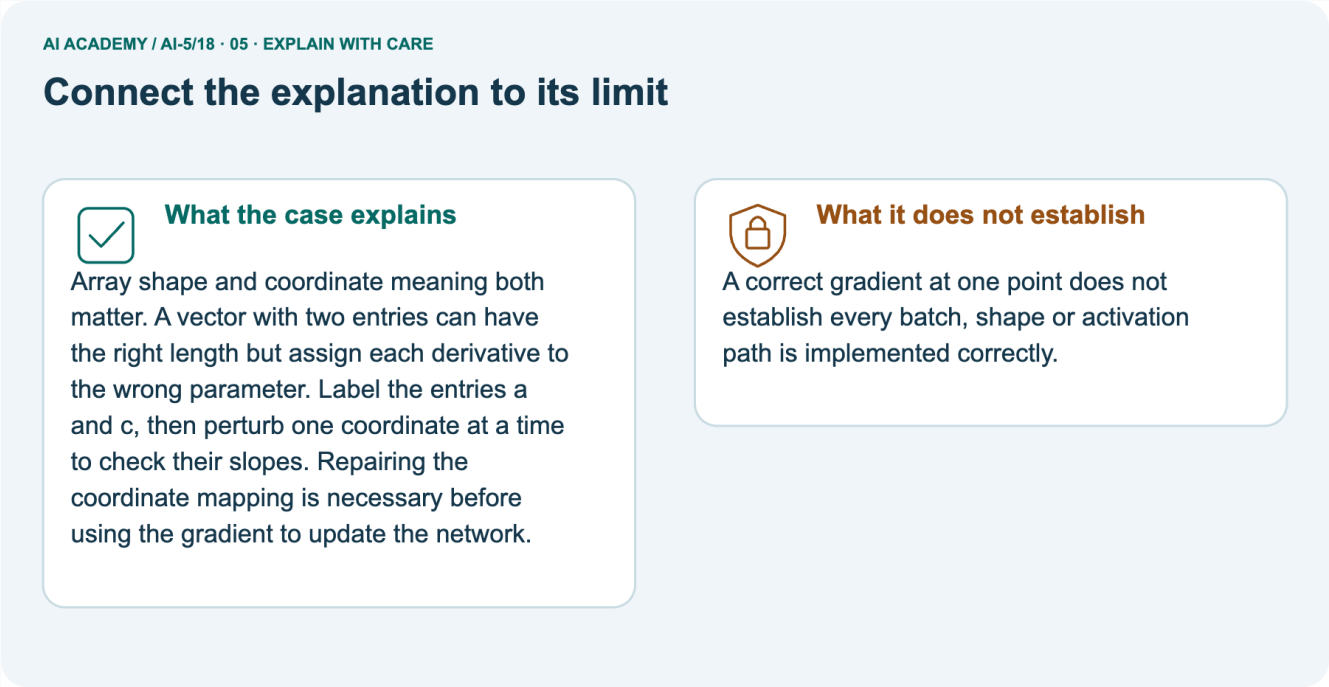
Original case: At rate 0.01: new [a, c]=[1.08, 0.12]; y = 2.52, loss 2.1904. Changed condition: The program changes its declared order to [c, a] and returns [-12,-8].

Expected result: That vector now matches the declared coordinate order. Reason: Gradient values are meaningful only when their parameter names and array positions agree.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### My picture explanation, deciding evidence and one limit

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Array **shape** and coordinate meaning both matter. A vector with two entries can have the right length but assign each derivative to the wrong parameter.

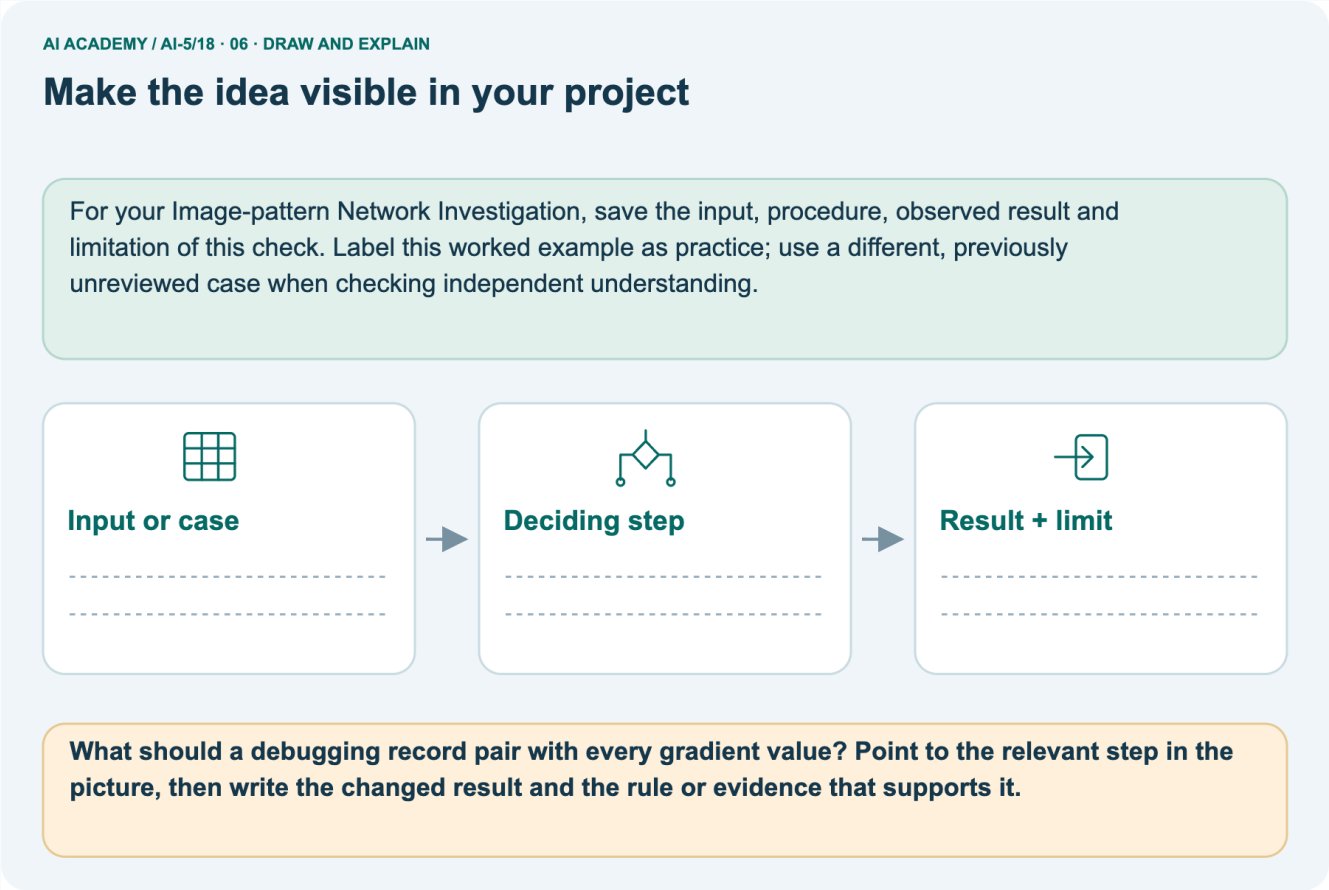
Label the entries a and c, then perturb one coordinate at a time to check their slopes. Repairing the coordinate mapping is necessary before using the gradient to update the network.

Limit: A correct gradient at one point does not establish every batch, **shape** or activation path is implemented correctly.

Explain the case, then point to a limit on the claim.

### My picture explanation, deciding evidence and one limit

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should a debugging record pair with every gradient value? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

**My picture explanation, deciding evidence and one limit**

▼ Earlier wording, alternatives and picture transcripts

**01. Debugging shapes and gradients: the case**

**02. Follow the deciding steps**

**03. Repair from the first disagreement**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.